# Module 10: Cross-Program Invocations (CPIs)

## Program Composition & Interoperability

# What are Cross-Program Invocations?

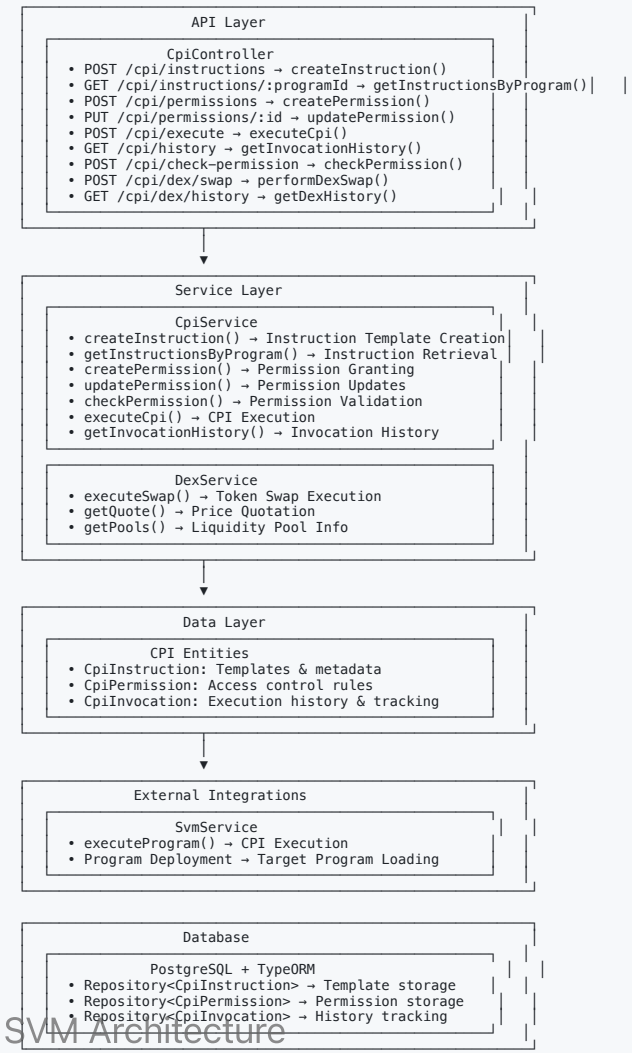
## CPI Fundamentals

- **Program Calls**: One program invoking another program's instructions
- **Composability**: Building complex functionality from simple programs
- **Trustless Execution**: Programs execute with full SVM runtime security
- **Atomic Operations**: All CPI calls succeed or fail together

## SVM Advantages

- **Single Transaction**: Multiple program interactions in one atomic operation
- **Shared State**: Programs can read and modify shared accounts
- **Gas Efficiency**: No additional transaction overhead for CPI calls
- **Security Isolation**: Each program maintains its own security boundaries

# Architecture Overview



# CPI Permission System

## Permission Management

```
interface CpiPermission {
  id: string;
  programId: string;          // Target program being granted access
  granterProgramId: string; // Program granting the permission
  permissionType: 'read' | 'write' | 'execute';
  accountId?: string;         // Specific account (optional)
  expiresAt?: Date;           // Permission expiration
  isActive: boolean;
}
```

## Permission Validation

```
async checkPermission(
  callerProgramId: string,
  targetProgramId: string,
  permissionType: string,
```

# Instruction Templates

## CPI Instruction Structure

```
interface CpiInstruction {
  id: string;
  programId: string;            // Target program ID
  callerProgramId: string;      // Calling program ID
  instructionData: any;         // Instruction payload
  accounts: AccountMeta[];      // Account metadata array
  methodName: string;           // Method identifier
  requiresPermission: boolean; // Permission check required
  permissionLevel: string;      // Required permission type
  isActive: boolean;
}

interface AccountMeta {
  pubkey: PublicKey;
  isSigner: boolean;
  isWritable: boolean;
}
```

# CPI Execution Flow

## Execution Process

```
async executeCpi(executionDto: ExecuteCpiDto): Promise<CpiInvocation> {
  // 1. Permission Check
  if (executionDto.requiresPermission) {
    const hasPermission = await this.checkPermission(
      executionDto.callerProgramId,
      executionDto.targetProgramId,
      executionDto.permissionType
    );

    if (!hasPermission) {
      throw new ForbiddenException('Insufficient CPI permissions');
    }
  }

  // 2. Create invocation record
  const invocation = await this.createInvocationRecord(executionDto);

  try {
    // 3. Execute via SVM service
    const result = await this.svmService.executeProgram({
      programId: executionDto.targetProgramId,
      instruction: executionDto.instruction,
      accounts: executionDto.accounts
    });

    // 4. Update invocation with success
    invocation.status = 'success';
    invocation.transactionId = result.signature;
    invocation.gasUsed = result.computeUnits;

  } catch (error) {
    // 5. Record failure
    invocation.status = 'failed';
    invocation.errorMessage = error.message;
  }
```

# DEX Operations

## Token Swap Implementation

```
async performDexSwap(swapDto: DexSwapDto): Promise<SwapResult> {
  // Get quote from DEX
  const quote = await this.getQuote({
    inputMint: swapDto.inputMint,
    outputMint: swapDto.outputMint,
    amount: swapDto.amount,
    slippage: swapDto.slippage
  });

  // Check slippage tolerance
  if (quote.priceImpact > swapDto.maxSlippage) {
    throw new BadRequestException('Price impact too high');
  }

  // Execute swap via CPI
  const swapInstruction = await this.createSwapInstruction(quote);

  const result = await this.cpiService.executeCpi({
    callerProgramId: this.dexProgramId,
    targetProgramId: quote.ammProgramId,
    instruction: swapInstruction,
    accounts: quote.accounts,
    requiresPermission: true,
    permissionType: 'execute'
  });

  return {
    signature: result.transactionId,
    outputAmount: quote.outputAmount,
    priceImpact: quote.priceImpact
```

# Common CPI Patterns

## Token Operations

```
// SPL Token Transfer via CPI
const tokenTransferIx = Token.createTransferInstruction(
  TOKEN_PROGRAM_ID,
  sourceTokenAccount,
  destinationTokenAccount,
  owner,
  [],
  amount
);

// Execute via CPI
await cpiService.executeCpi({
  callerProgramId: myProgramId,
  targetProgramId: TOKEN_PROGRAM_ID,
  instruction: tokenTransferIx,
  accounts: [
    { pubkey: sourceTokenAccount, isSigner: false, isWritable: true },
    { pubkey: destinationTokenAccount, isSigner: false, isWritable: true },
    { pubkey: owner, isSigner: true, isWritable: false }
```

# Security Features

## CPI Security Measures

- **Permission Validation**: Granular access control enforcement
- **Account Verification**: Address validation and ownership checks
- **Execution Isolation**: Failure containment within CPI boundaries
- **Audit Trail**: Complete invocation logging and tracking
- **Gas Metering**: Resource usage monitoring and limits

## Error Handling

```
class CpiExecutionError extends Error {
  constructor(
    public readonly invocationId: string,
    public readonly errorCode: string,
    public readonly details: any
  ) {
```

# API Endpoints

## Instruction Management

- `POST /cpi/instructions` - Create CPI instruction template
- `GET /cpi/instructions/:programId` - Get instructions for program

## Permission Management

- `POST /cpi/permissions` - Create CPI permission
- `PUT /cpi/permissions/:id` - Update permission
- `POST /cpi/check-permission` - Validate permission

## Execution & History

- `POST /cpi/execute` - Execute CPI call
- `GET /cpi/history` - Get invocation history

# Advanced CPI Patterns

## Program Derived Address Signing

```
// CPI with PDA signing
const [pdaAddress, bump] = await PublicKey.findProgramAddress(
  [Buffer.from('authority')],
  programId
);

const instruction = new TransactionInstruction({
  keys: [
    { pubkey: pdaAddress, isSigner: false, isWritable: true },
    // ... other accounts
  ],
  programId: targetProgramId,
  data: instructionData
});

// Execute with PDA authority
await cpiService.executeCpi({
  callerProgramId: programId,
  targetProgramId: targetProgramId,
  instruction,
  accounts: instruction.keys,
```

# Key Takeaways

## CPI Architecture Benefits

- **Program Composability**: Build complex protocols from simple programs
- **Atomic Execution**: All CPI calls succeed or fail together
- **Security Isolation**: Each program maintains its own security model
- **Performance Efficiency**: No additional transaction overhead

## SVM CPI Advantages

- **High Throughput**: Parallel CPI execution across programs
- **Shared State**: Programs can securely share and modify accounts
- **Gas Optimization**: Efficient resource usage for complex operations
- **Trustless Composition**: Programs can interact without trusted intermediaries